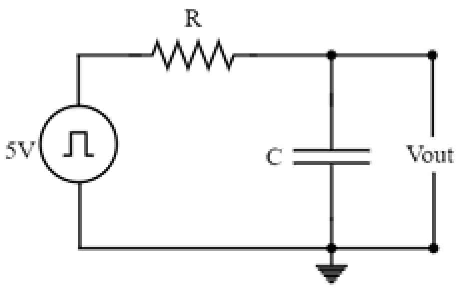
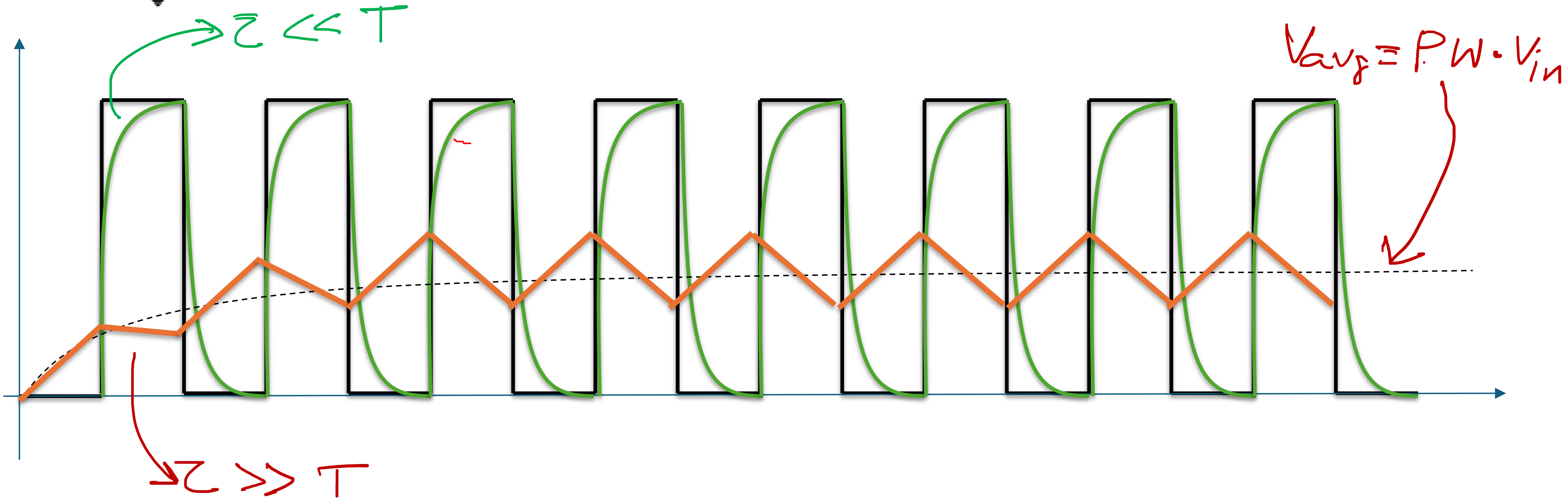
SI-2026 Analog IC Design Review: Circuits

28-29 May 2026

2. Draw the output wave form of the below figure for a pulse input.
   a) RC << T
   b) RC >> T
   c) Find the average output voltage for the above two cases.



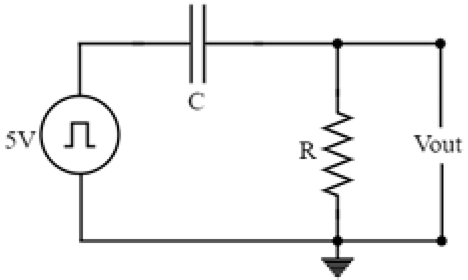


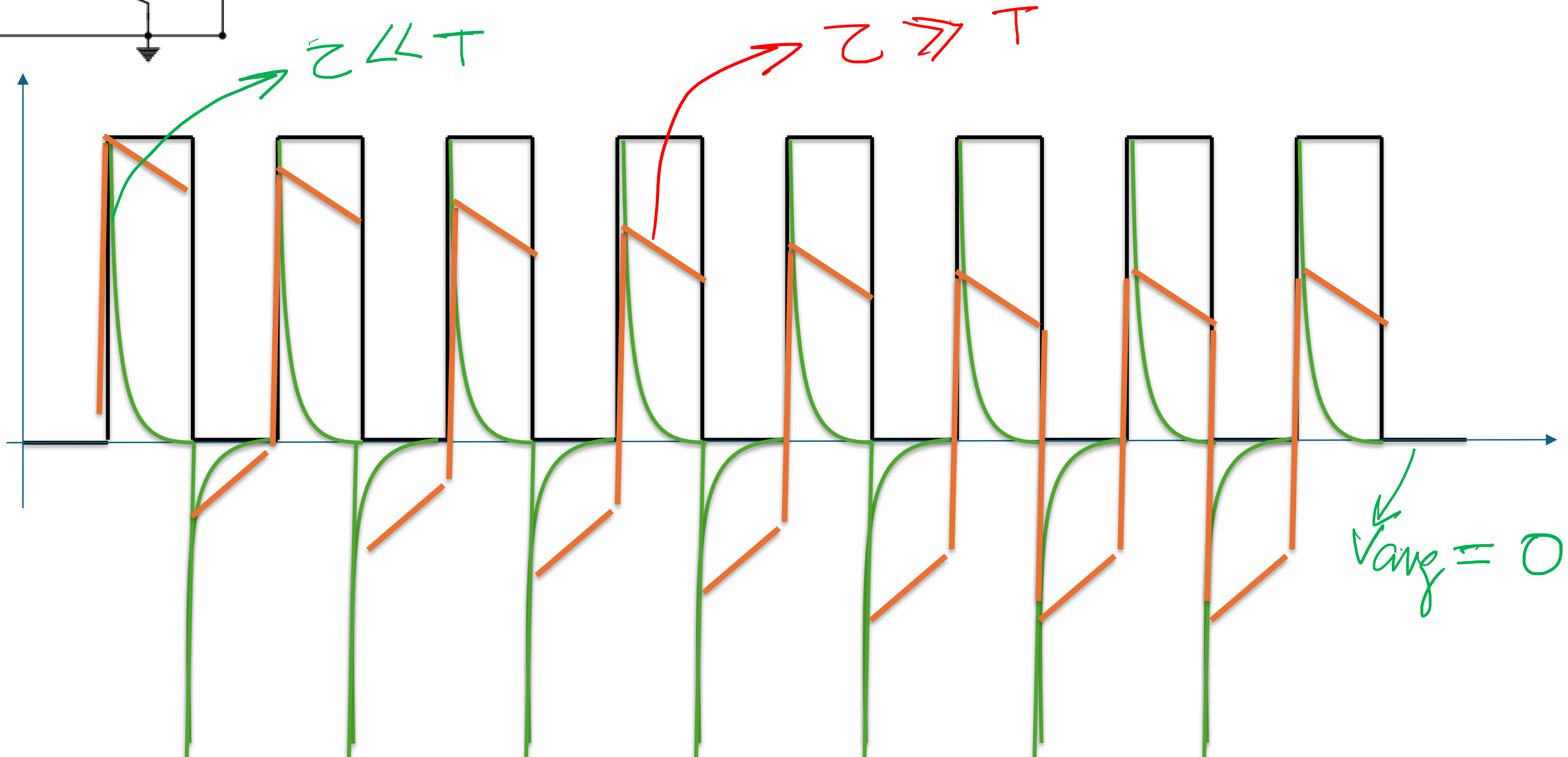
4. Draw the output wave form of the below figure for a pulse input.

   d) RC << T

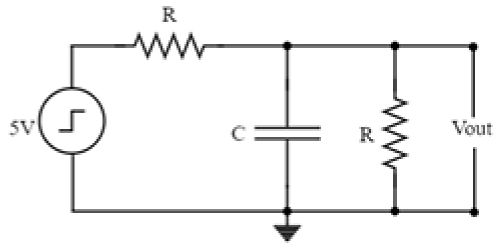
   e) RC >> T

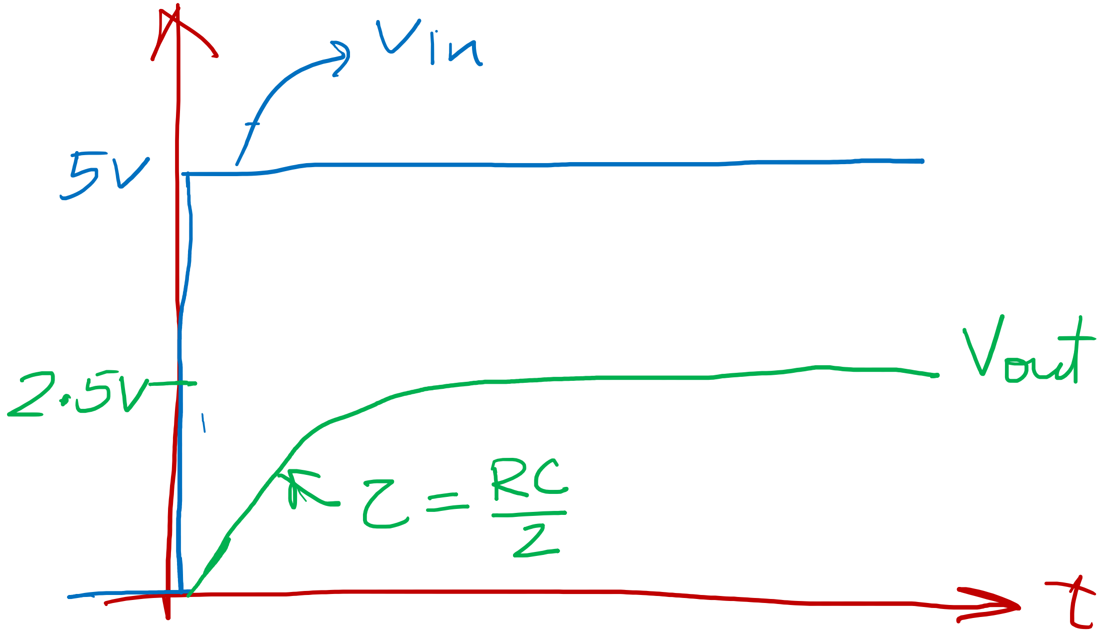
   f) Find the average output voltage for the above two cases.





5. Draw the step response of the below circuit diagram.





① Determine the order: 1^{st} order

② Type: RC (low pass) or CR (HPF): RC/LPF

③ Initial value: 0

④ Final value: $V_{DD}/2 = 2.5V$

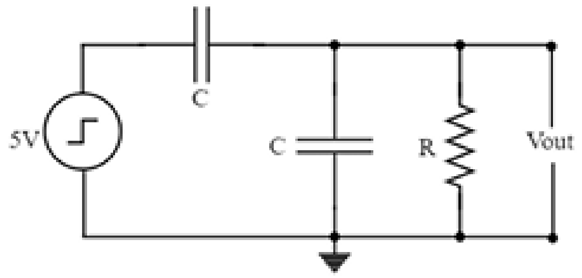
⑤ τ ? $= R_{eq} C = R/2 \cdot C$

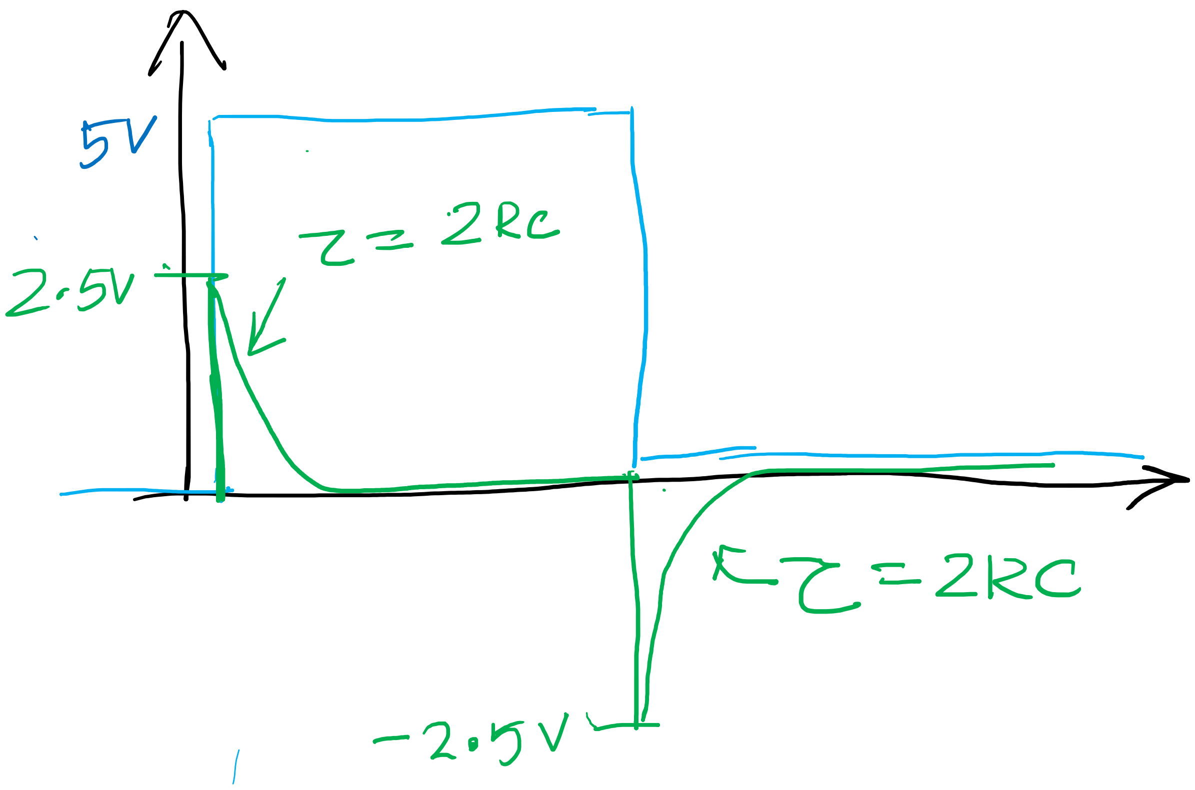
↳ Follow open-ckt time constant

① All independent srcs $= 0$

② R_{eq} in Parallel to C

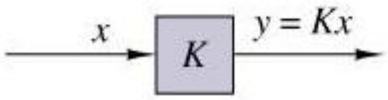
6. Draw the step response of the below circuit diagram.



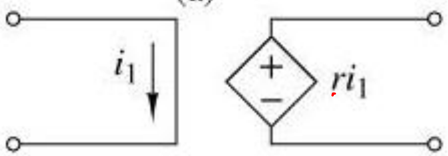


Active Circuits

• Linear Dependent Sources

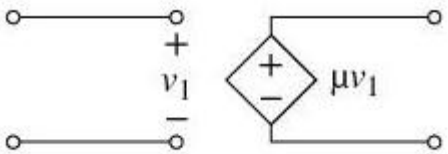


(a)



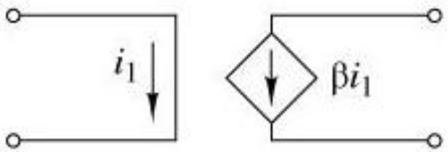
(b) CCVS

Transimpedance amplifier



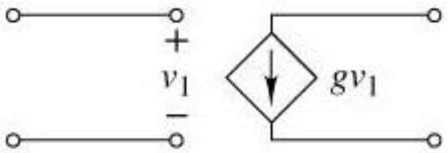
(c) VCVS

Voltage amplifier (OpAmp can be modeled w/ this)



(d) CCCS

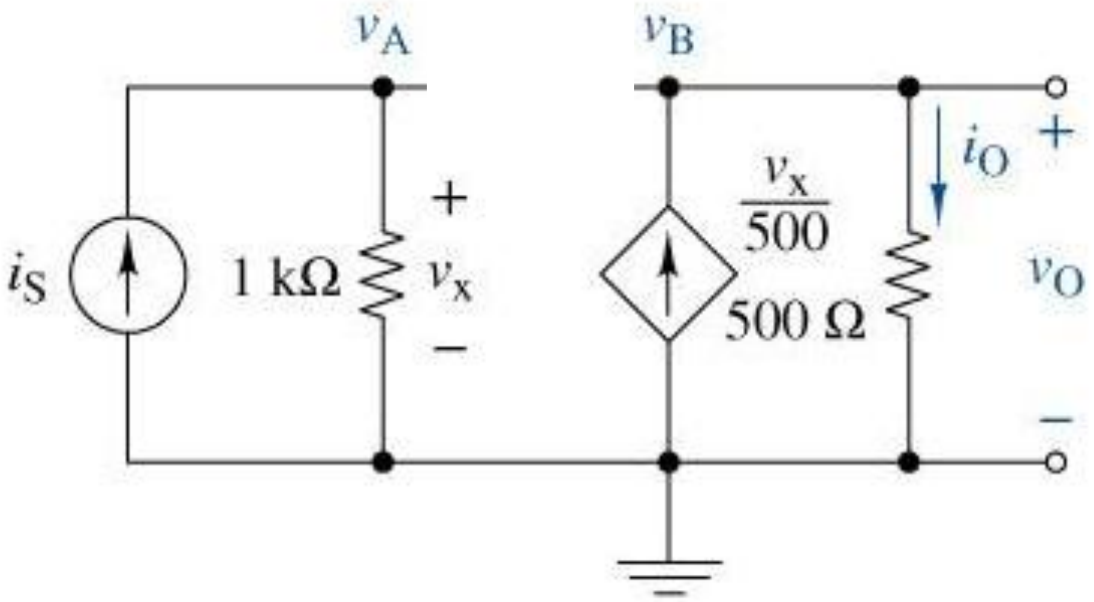
Current amplifier



(e) VCCS

Transconductance amplifier (OTAs can be modeled)

- **Analysis using Dependent Sources**



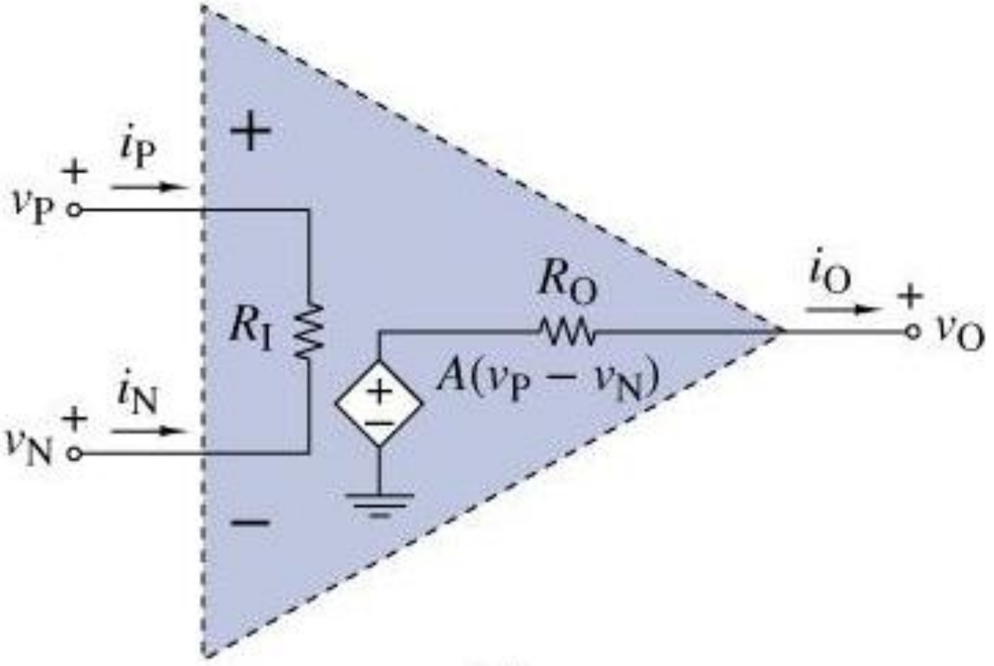
$$v_o = i_o \cdot r_o$$

$$= \frac{v_x}{\cancel{500}} \times \cancel{500}$$

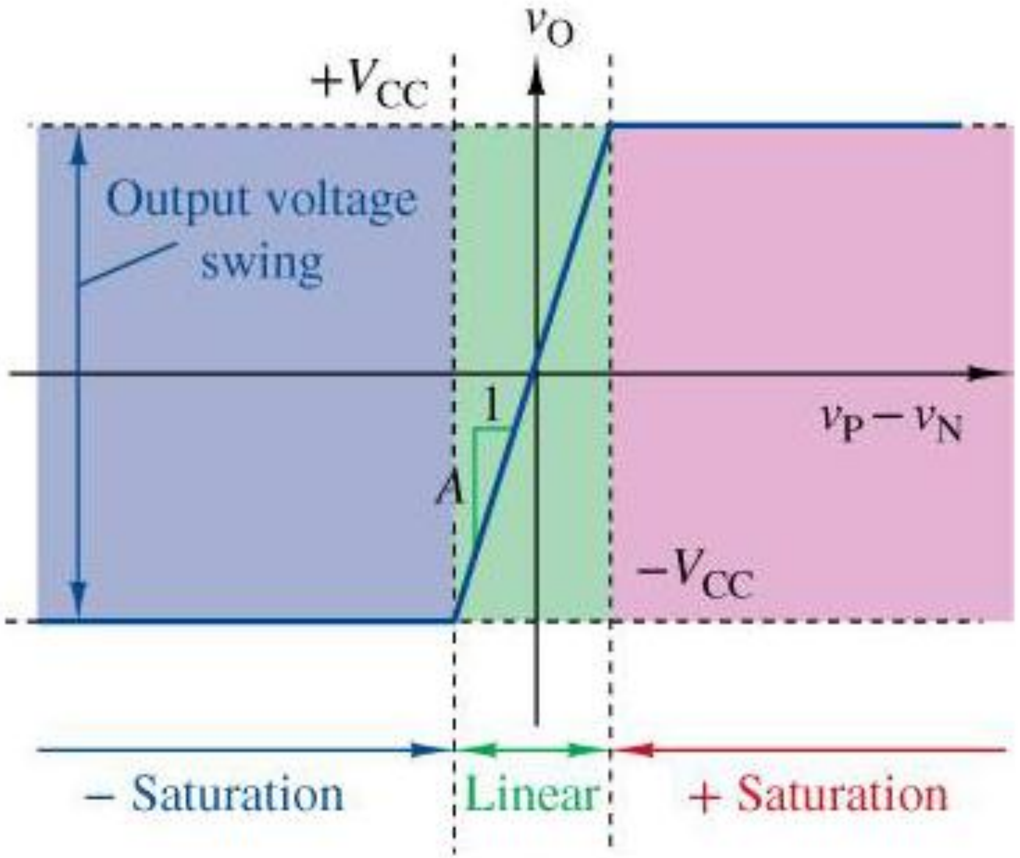
$$= i_s \cdot 1\,k\Omega$$

$$\boxed{\frac{v_o}{i_s} = 1\,k\Omega}$$

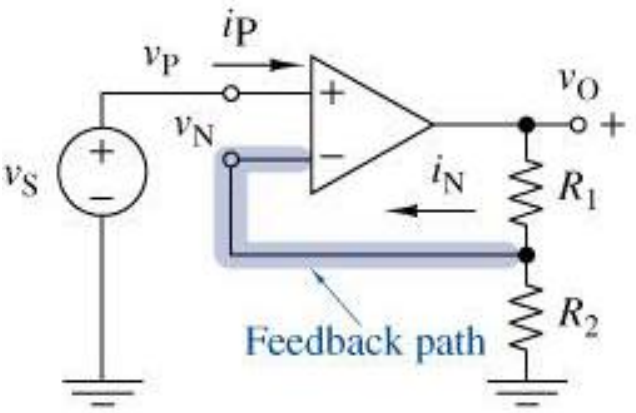
• Operational Amplifier Model

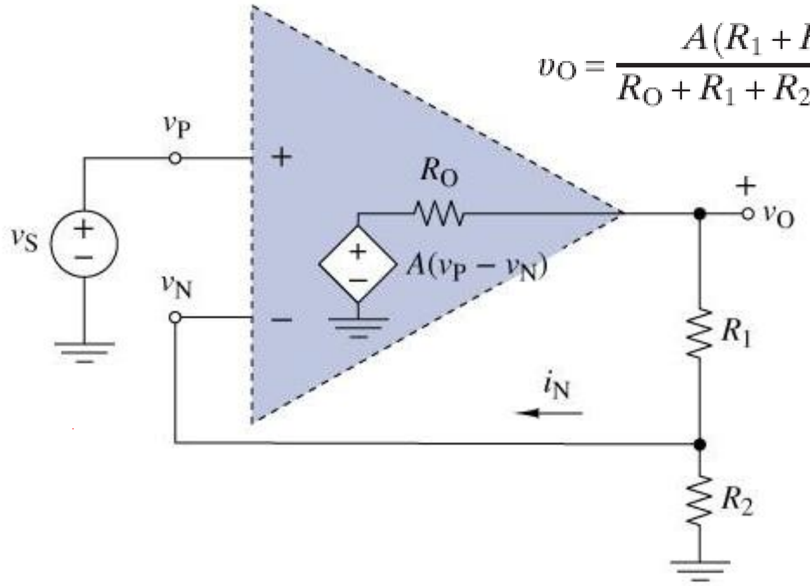


(b)



- **Operational Amplifier Analysis**



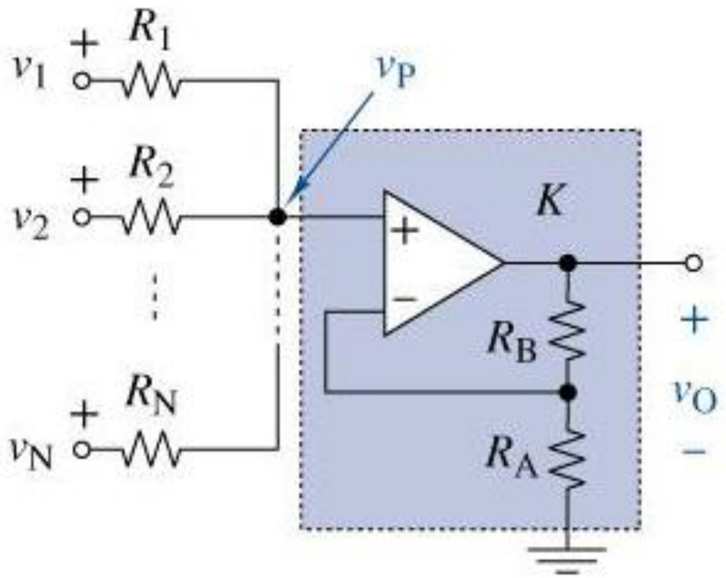


$$v_O = \frac{A(R_1 + R_2)}{R_O + R_1 + R_2(1 + A)} v_S$$

In ideal case $R_O = 0$ & $A = \infty$

$$\frac{v_o}{v_s} = \frac{R_1 + R_2}{\frac{R_O}{A} + \frac{R_1}{A} + R_2\left(\frac{1}{A} + 1\right)} = 1 + \frac{R_1}{R_2}$$

- **Operational Amplifier Analysis**

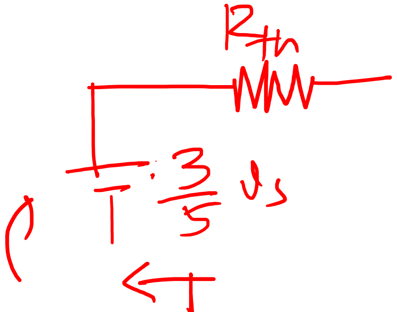


$$\frac{v_o}{v_p} = 1 + \frac{R_B}{R_A}$$

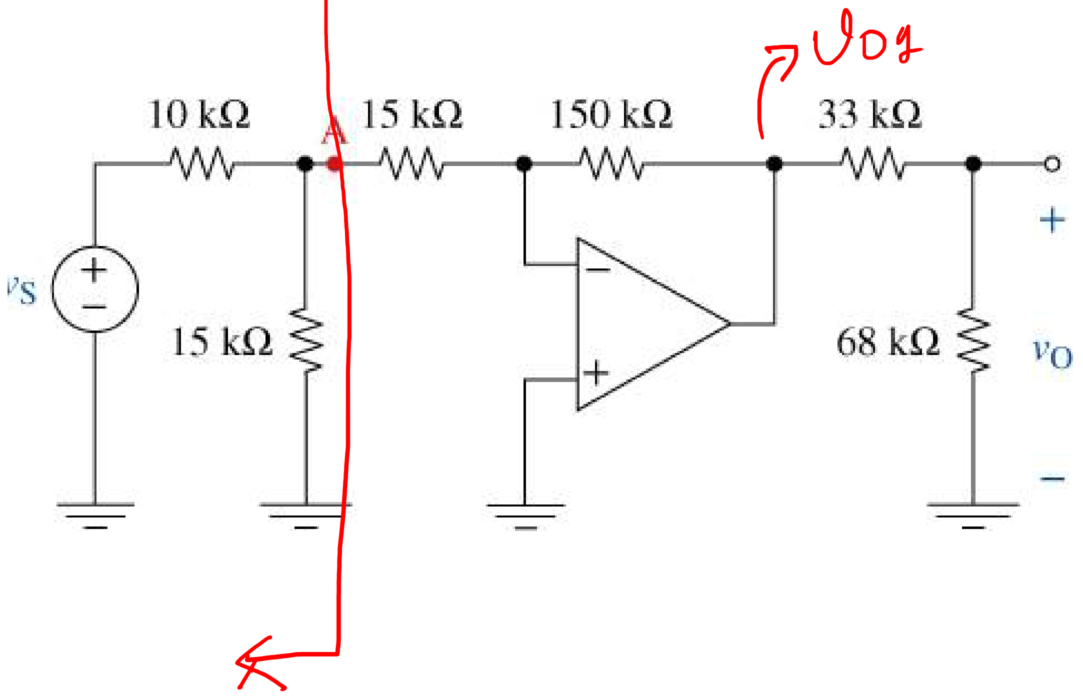
$$\frac{v_p}{v_1} = \frac{R_2 \parallel R_3 \parallel \ldots \parallel R_N}{\sum R_K}$$

$\lll y$: $$\frac{v_p}{v_2} = \frac{R_1 \parallel R_3 \parallel \ldots \parallel R_N}{\sum R_K}$$

By superposition : $\left(\frac{v_o}{v_1} + \frac{v_o}{v_2} + \ldots\right)$

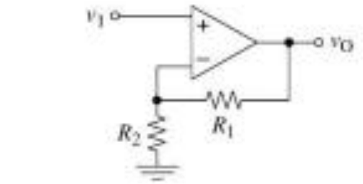
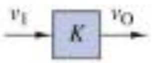
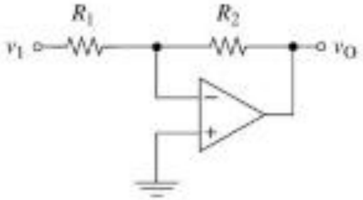
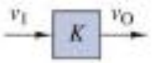
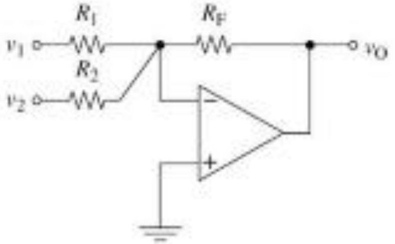
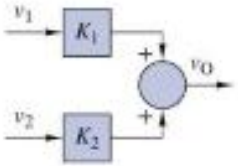
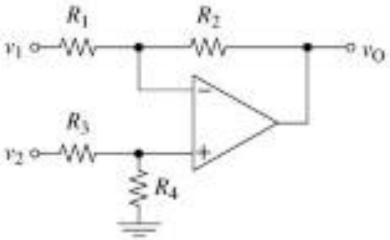
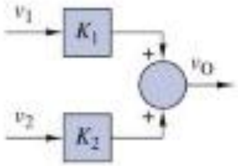
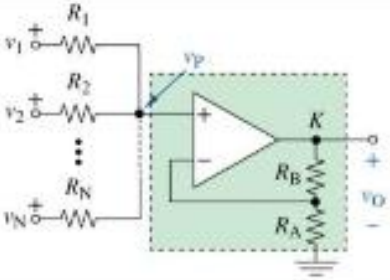
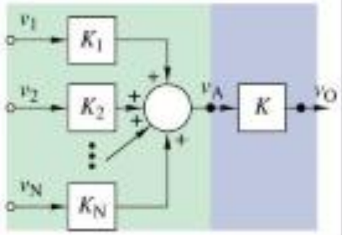


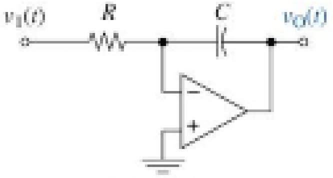
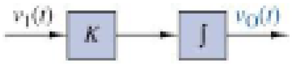
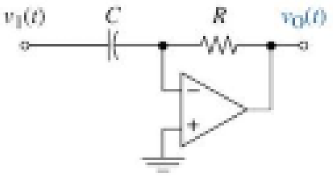
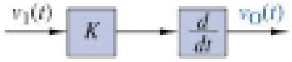
$$\frac{v_o}{v_s} = \frac{3}{5} \times -\frac{150k}{21k} \times \frac{68k}{(68+33)k} = -\frac{30}{7} \times \frac{68}{101} = -2.88$$



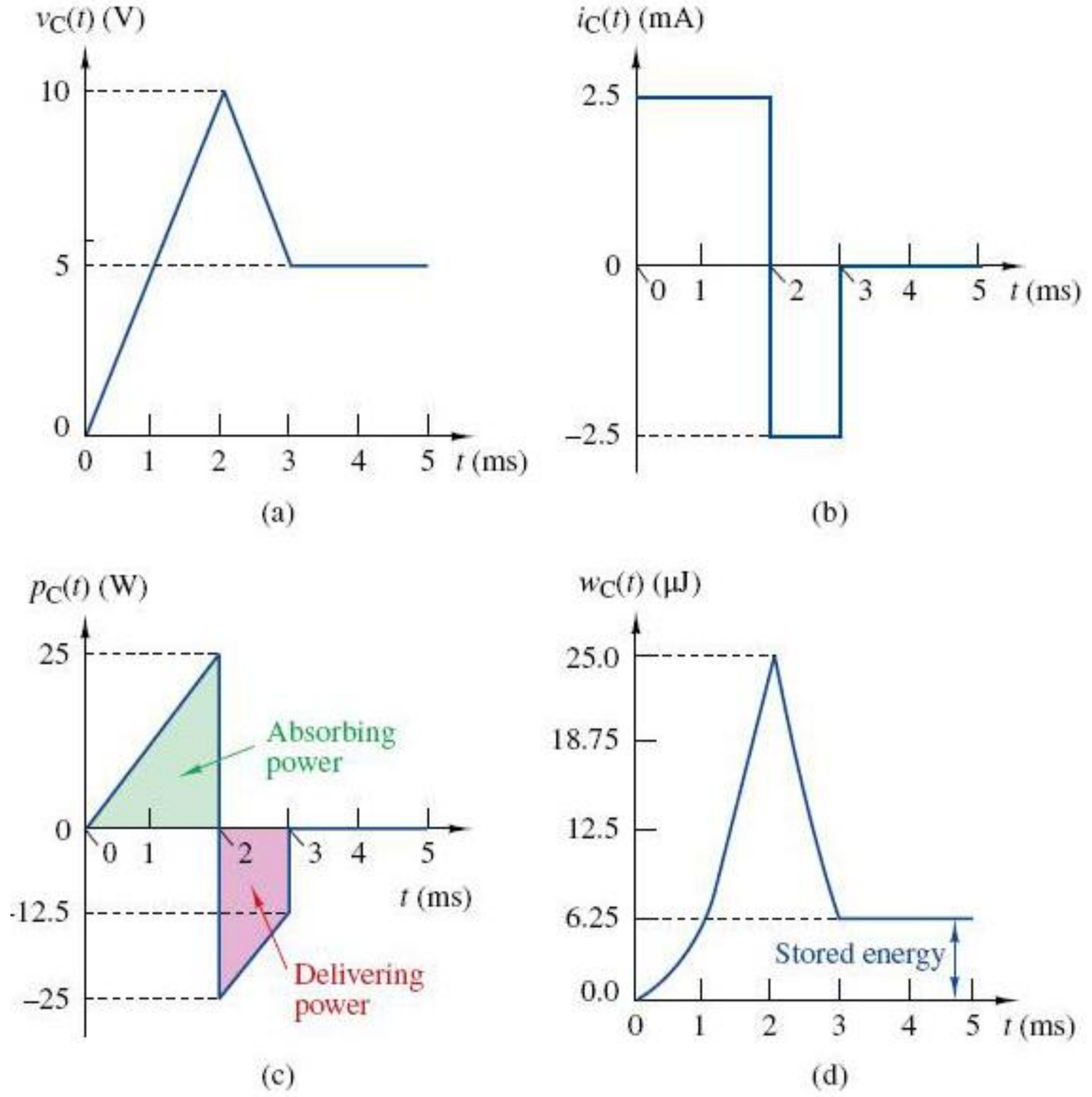
Thevenin equn:

$$V_{TH} = v_s \times \frac{15}{25} = \frac{3}{5} v_s \quad \Bigg| \quad R_{TH} = \frac{15k \times 10k}{15k + 10k} = 6\ k\Omega$$

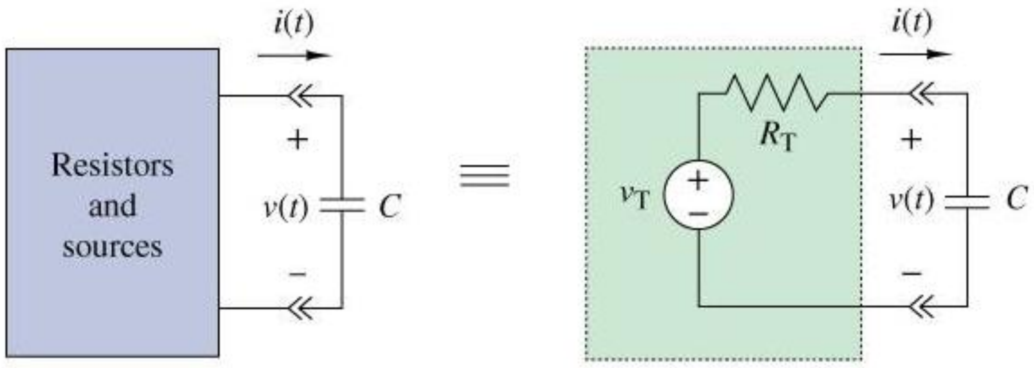
Circuit	Block diagram	Gains
	Non Inverting 	$K = \frac{R_1 + R_2}{R_2}$
	Inverting 	$K = -\frac{R_2}{R_1}$
	Inverting Summer 	$K_1 = -\frac{R_F}{R_1}$ $K_2 = -\frac{R_F}{R_2}$
	Subtractor 	$K_1 = -\frac{R_2}{R_1}$ $K_2 = \left(\frac{R_1 + R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)$
	Non-Inverting Summer 	$K = \frac{R_B + R_A}{R_A}$ $K_1 = \frac{(R_2 \parallel R_3 \parallel \cdots R_N)}{R_1 + (R_2 \parallel R_3 \parallel \cdots R_N)}$ $K_N = \frac{(R_1 \parallel R_2 \parallel \cdots R_{N-1})}{R_N + (R_1 \parallel R_2 \parallel \cdots R_{N-1})}$

 Integrator		$K = -\frac{1}{RC}$
 Differentiator		$K = -RC$

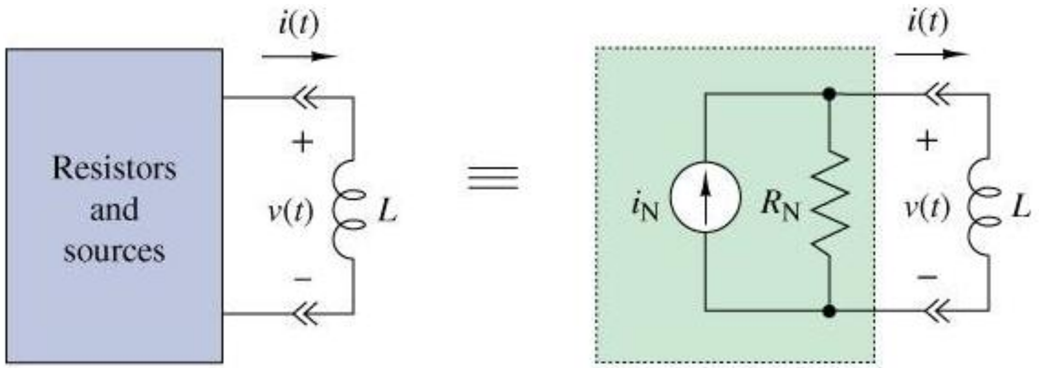
- **Capacitance**



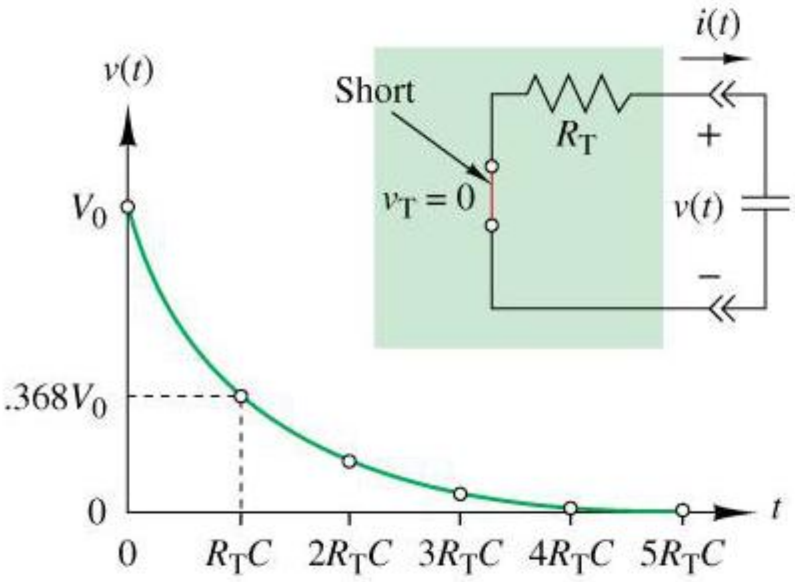
• First-Order Circuits



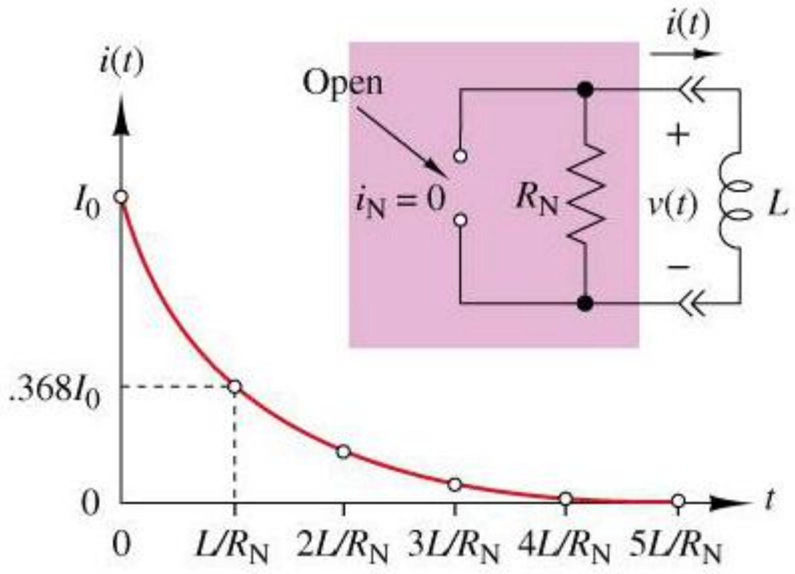
(a)



(b)

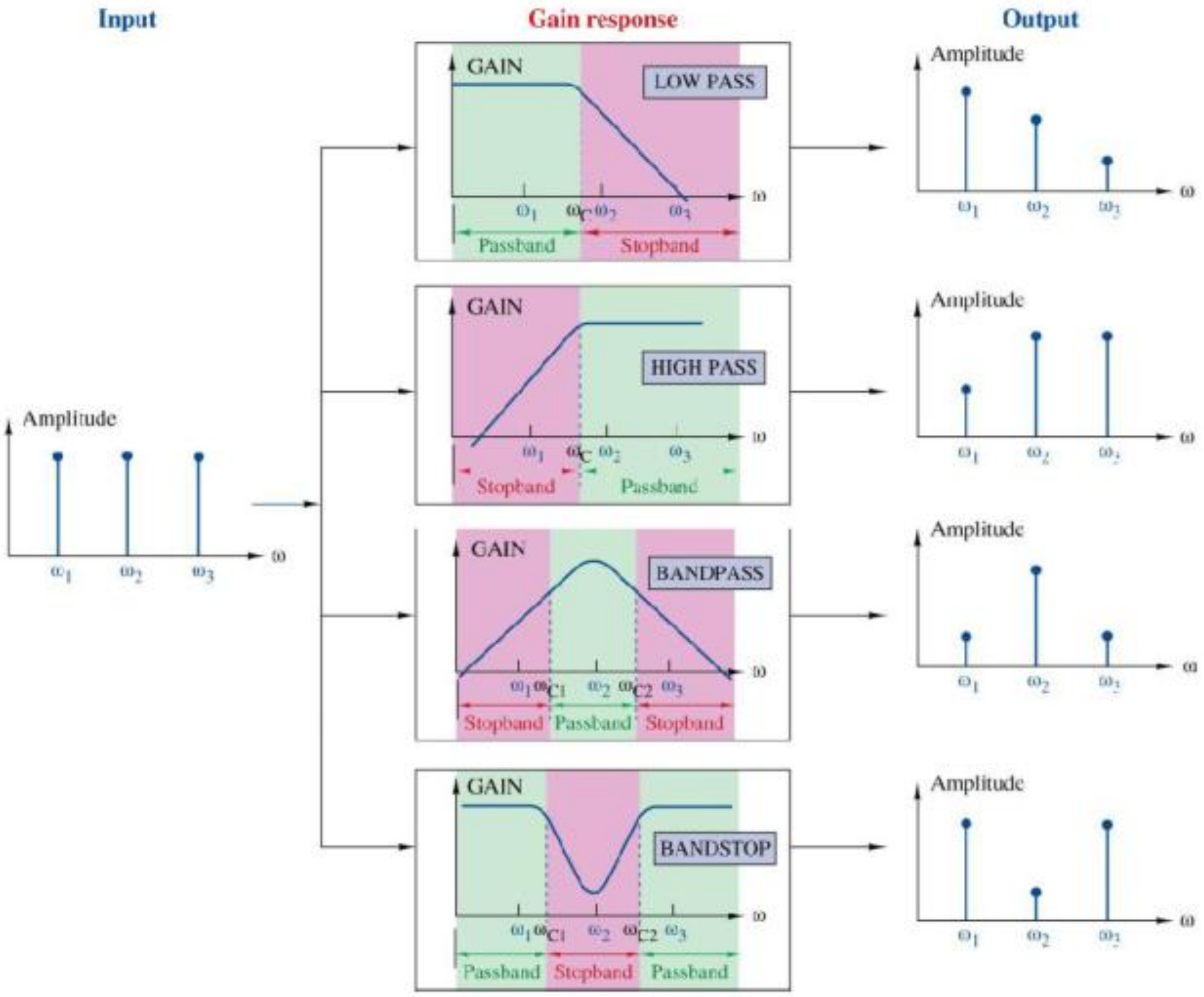


(a)



(b)

• **Frequency Response (Four Basic Gain Responses)**



- **First-Order Low-Pass Response**

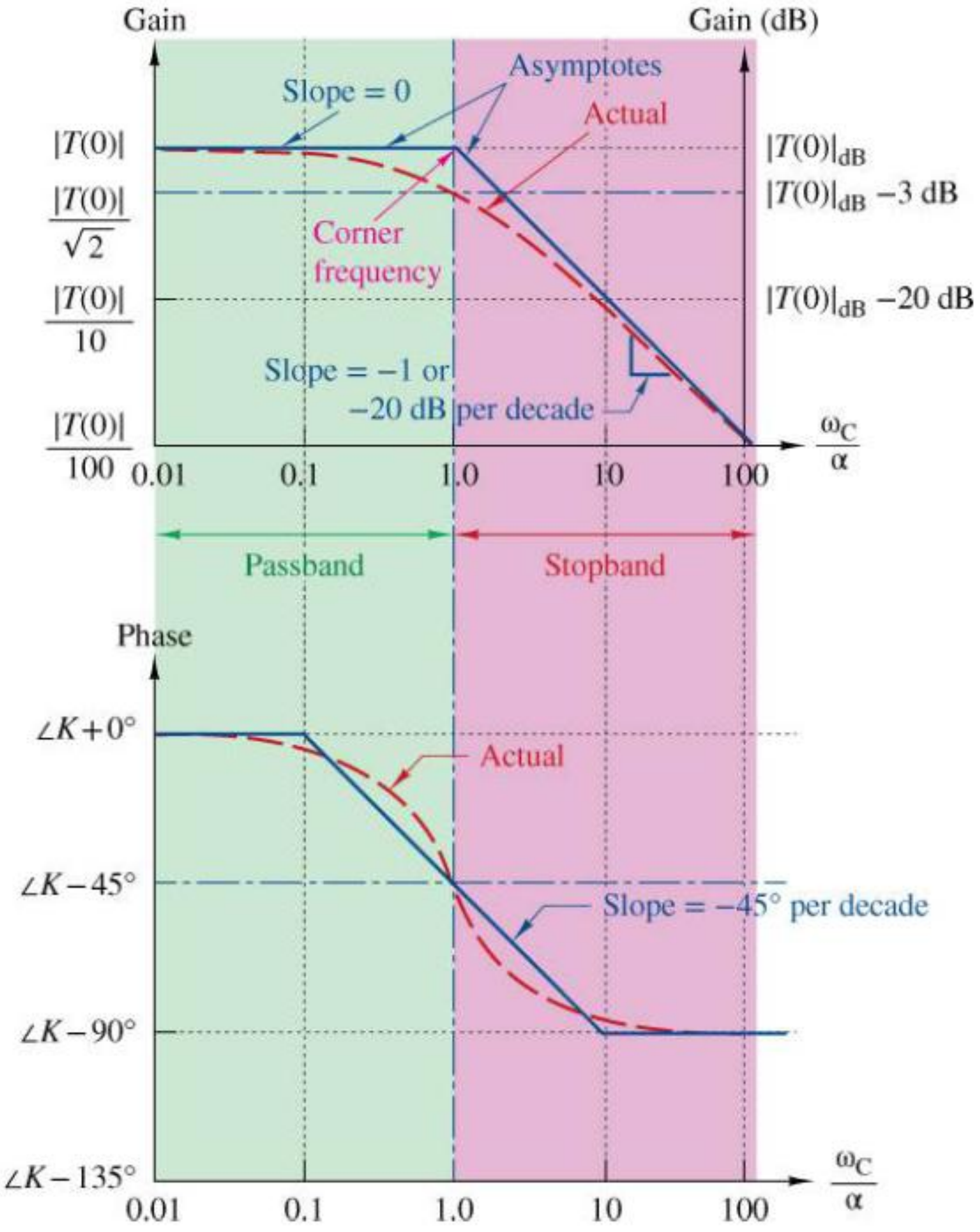
$$T(s) = \frac{K}{s + \alpha}$$

$$T(j\omega) = \frac{K}{j\omega + \alpha}$$

$$|T(j\omega)| = \frac{|K|}{\sqrt{\omega^2 + \alpha^2}}$$

$$\theta(\omega) = angle(K) - \tan^{-1}(\omega/\alpha)$$

$\lvert T(j\omega)\rvert$	$\lvert T(j\omega)\rvert_{dB}$
10^{3}	60
10^{2}	40
10	20
2	6
$\sqrt{2}$	3
1	0
$1/\sqrt{2}$	−3
0.5	−6
10^{-1}	−20
10^{-2}	−40
10^{-3}	−60



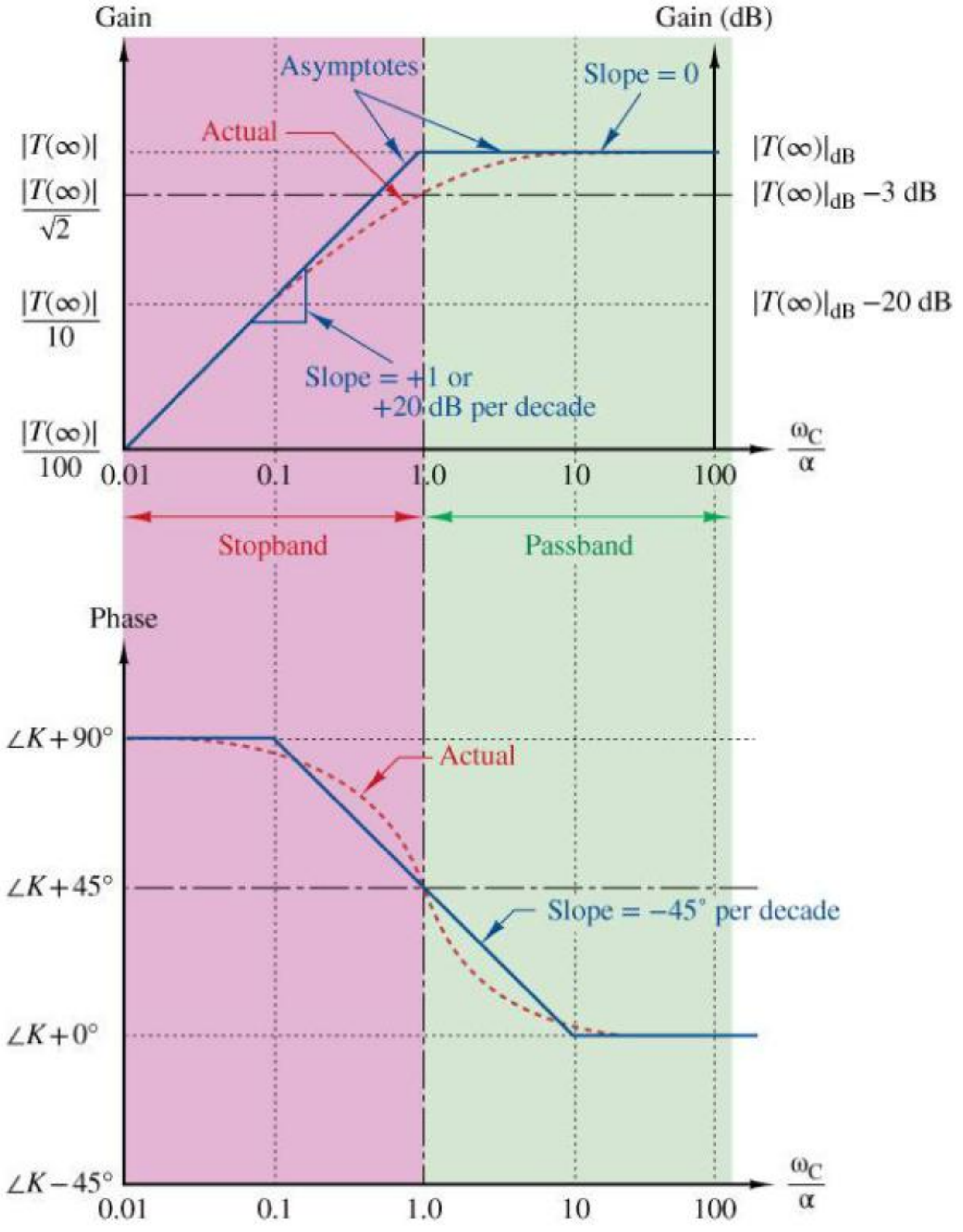
- **First-Order High-Pass Response**

$$T(s) = \frac{Ks}{s+\alpha}$$

$$T(j\omega) = \frac{j\omega K}{j\omega + \alpha}$$

$$|T(j\omega)| = \frac{|K|\omega}{\sqrt{\omega^2 + \alpha^2}}$$

$$\theta(\omega) = angle(K) + 90^\circ - \tan^{-1}(\omega/\alpha)$$

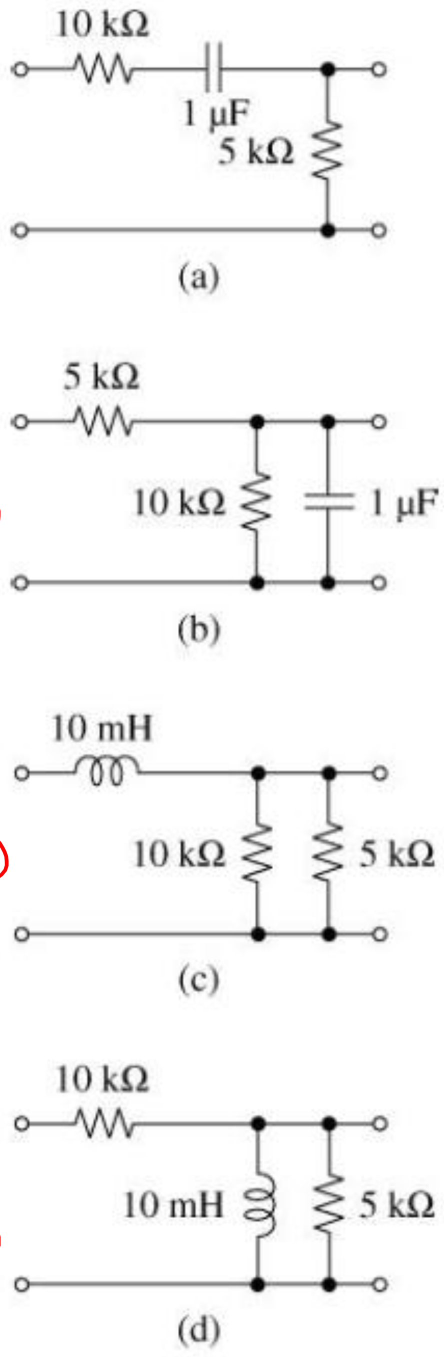


(a)

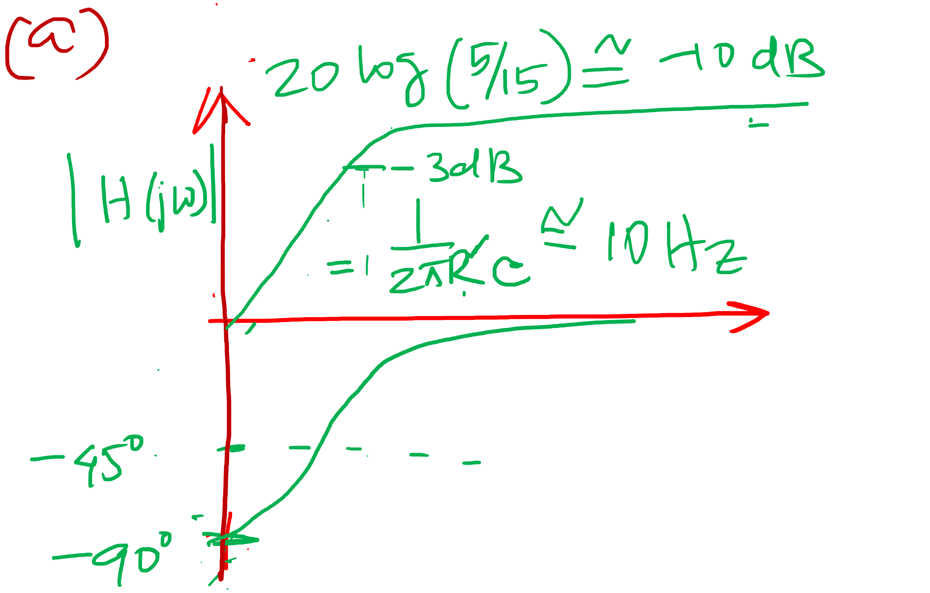
(b)

(c)

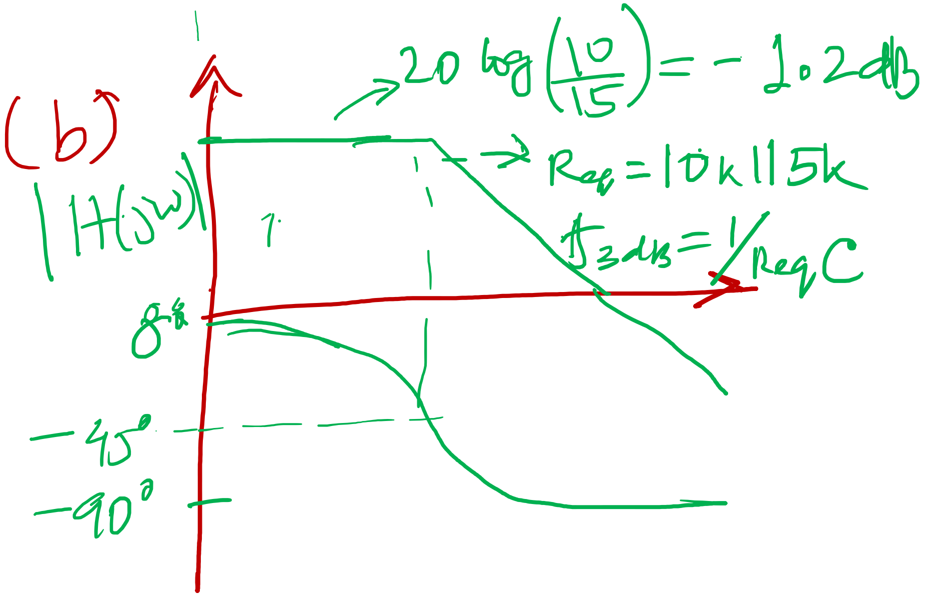
(d)

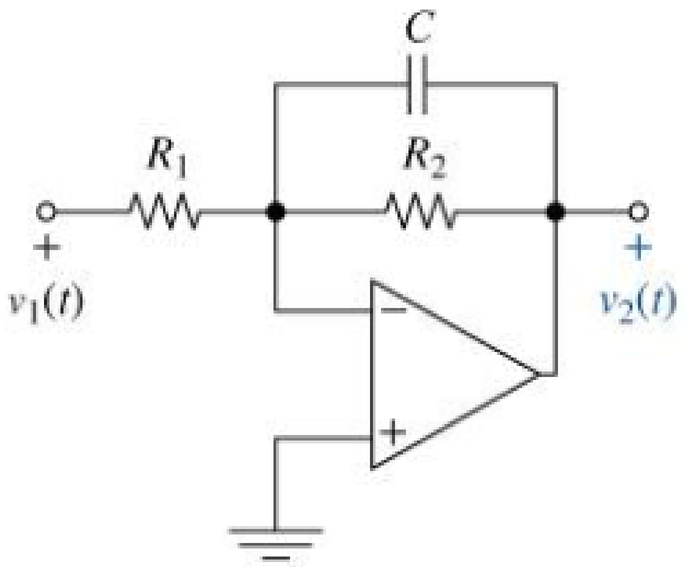

10 kΩ
1 μF
5 kΩ
(a)
5 kΩ
10 kΩ
1 μF
(b)
10 mH
10 kΩ
5 kΩ
(c)
10 kΩ
10 mH
5 kΩ
(d)

(a)


20 log (5/15) ≃ -10 dB
|H(jω)|
-3dB
$= \frac{1}{2\pi RC} \simeq 10\,Hz$
-45°
-90°

(b)


$20 \log\left(\frac{10}{15}\right) = -1.2\,dB$
$R_{eq} = 10k \| 5k$
$f_{3dB} = 1/R_{eq}C$
|H(jω)|
-45°
-90°

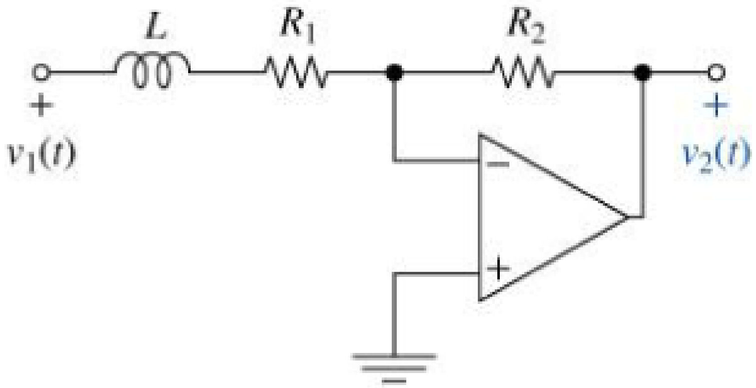


(a)

→ LPF;

Gain $= -\frac{R_2}{R_1}$

pole $= \frac{1}{2\pi R_2 C}$



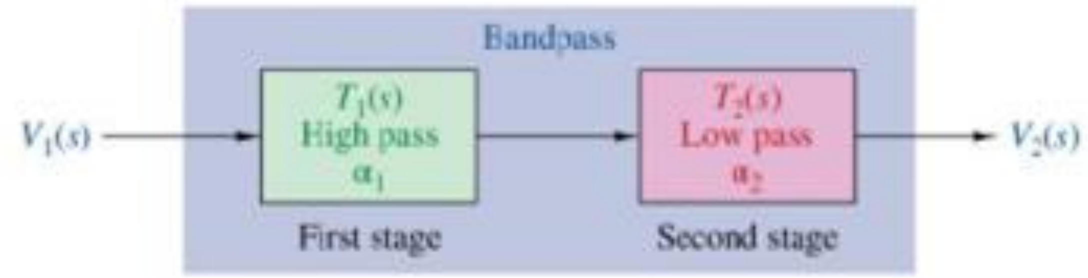
(b)

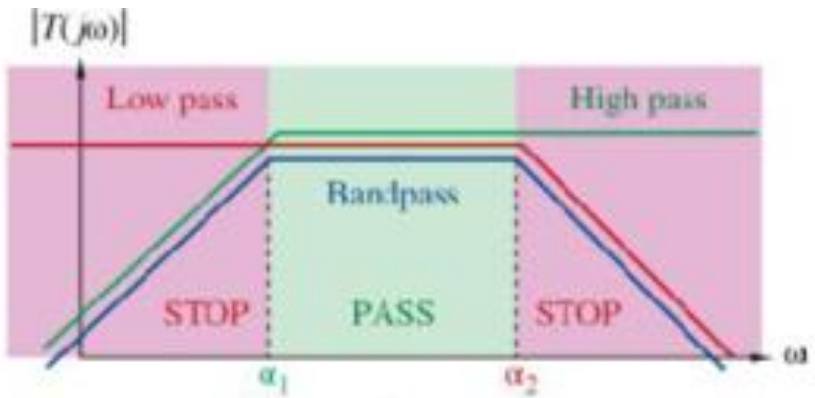
LPF

Gain $= -\frac{R_2}{R_1}$

pole $= \frac{L}{R_1}$

• First-Order Band-Pass Response





$$T(s) = T_1(s) \times T_2(s) = \underbrace{\left(\frac{K_1 s}{s+\alpha_1}\right)}_{\text{high pass}} \underbrace{\left(\frac{K_2}{s+\alpha_2}\right)}_{\text{low pass}}$$

$$|T(j\omega)| = \underbrace{\left(\frac{|K_1|\omega}{\sqrt{\omega^2+\alpha_1^2}}\right)}_{\text{high pass}} \underbrace{\left(\frac{|K_2|}{\sqrt{\omega^2+\alpha_2^2}}\right)}_{\text{low pass}}$$

Low frequency ($\omega \ll \alpha_1 \ll \alpha_2$): $$|T(j\omega)| \to \underbrace{\left(\frac{|K_1|\omega}{\alpha_1}\right)}_{\text{high pass}} \underbrace{\left(\frac{|K_2|}{\alpha_2}\right)}_{\text{low pass}} = \frac{|K_1||K_2|\omega}{\alpha_1\alpha_2}$$

High frequency ($\alpha_1 \ll \alpha_2 \ll \omega$) : $$|T(j\omega)| \to \underbrace{\left(\frac{|K_1|\omega}{\omega}\right)}_{\text{high pass}} \underbrace{\left(\frac{|K_2|}{\omega}\right)}_{\text{low pass}} = \frac{|K_1||K_2|}{\omega}$$

Mid-frequency ($\alpha_1 \ll \omega \ll \alpha_2$): $$|T(j\omega)| \to \underbrace{\left(\frac{|K_1|\omega}{\omega}\right)}_{\text{high pass}} \underbrace{\left(\frac{|K_2|}{\alpha_2}\right)}_{\text{low pass}} = \frac{|K_1||K_2|}{\alpha_2}$$